Volatility of Treasury Yields During Election Cycles

Alexander Tuosto

Pace University

FIN 325: Data Analysis in Finance

Author Note

This paper is based on independent research I conducted as part of a team project for Data Analysis in Finance. I served as lead researcher and designed the research question and project scope, wrote all the R code for the analysis pipeline, and was responsible for quality control on the final statistical work and presentation.

Abstract

This study tests whether U.S. presidential elections affect Treasury yield volatility, using daily FRED data from September 1981 through November 2025 across three maturities (3-month, 1-year, 10-year). Using two-sample t-tests, probit regressions, and GJR-GARCH(1,1) modeling, I find no significant election-year effect on extreme volatility for the 3-month or 1-year Treasury, and a significant *decrease* in extreme-volatility probability for the 10-year during election years, a finding contrary to the original hypothesis. These results suggest Treasury volatility is driven more by macroeconomic conditions and structural differences across maturities than by the electoral calendar, consistent with prior literature on political uncertainty and asset prices.

U.S. Treasury securities are widely considered the world's benchmark risk-free asset, as their yields indicate an interest rate which is (virtually) free of default risk (Liang, 2025). The cost for firms, as well as other fixed income market participants, to issue debt is largely based off Treasury yields because of their role as the benchmark rate. Ultimately, understanding what drives Treasury volatility matters not just to firms issuing debt but to anyone trying to gauge the stability of the broader economy.

General presidential elections warrant uncertainty, as it is difficult to account for irrational voter behavior in forecasts. Political parties differ in their views on fiscal policies such as corporate governance, trade policy, and taxation, all of which materially influence actions in fixed-income markets. In response to the uncertainty caused by presidential elections, participants in fixed-income markets may become more reactive to news and price this into treasury returns. I hypothesize that political uncertainty in years containing a presidential election will lead to a higher probability that Treasury securities' returns experience volatility greater than two standard deviations.

This is tested across three maturities (3-month, 1-year, and 10-year Treasuries) using daily data from the Federal Reserve Economic Database (FRED) spanning September 1981 through November 2025, a period covering eleven presidential elections. My hypothesis is that shorter-maturity instruments would show the clearest election-year volatility effect, since short-term yields are more sensitive to near-

term policy uncertainty, while longer-maturity yields would be insulated from this sensitivity, as they are largely driven by expected future interest rates and inflation expectations.

Literature Review

Previous literature explores both how Treasury markets behave as well as general financial market uncertainty caused by elections, but none specifically cover how Treasuries move in these uncertain times.

The article by Moench & Soofi-Siavash (2022) looks to identify what causes Treasury yields to move. Their research showed that "yield news shocks" or innovations that affect yields with a lag, can explain 50% of the variation when forecasting up to three years out. In this study, Treasury yields are separated into two main components: an expectation of future short-term rates and a term premium. The authors found that when a shock to yields materializes, it moves these two components in opposite directions, but that the term premium quickly returns to its pre-shock level while expected future rates fall more slowly. These shocks mostly result from stock market volatility and negative news about the business cycle; the Federal Reserve may respond to any negative news, which would also cause Treasury yields to fluctuate. These findings relate to this paper by revealing that Treasury yields are largely affected by news of economic downturns, which reinforces the idea to control for recession conditions. These findings also suggest that if elections act like news shocks, their effect on Treasury volatility may not show up immediately, but rather with a lag.

Pástor and Veronesi (2013) seek to quantify how uncertainty about future government actions affects market prices. They find that higher political uncertainty causes stocks to have higher correlation and more volatility. This study shows that political uncertainty is higher in poorer economic conditions, as the government is likely to take action to rectify them; the uncertainty comes from simply not knowing what actions the government will take. Findings also show that the effects of uncertainty regarding future government actions are larger in weaker economies, as investors respond more to political signals. The

effect of political uncertainty is large enough that the authors suggest it should have its own risk premium outside of regular economic risk. While Pástor and Veronesi measure stock market performance, the takeaways are still valuable to this paper. Political uncertainty does not have the same effect on asset prices throughout all economic states, which supports this paper's decision to control for recession conditions.

Goodell, McGee, and McGroarty (2020) provide research closest to the question in this paper, as they specifically look at seven U.S. presidential elections to determine how political uncertainty manifests itself in financial markets. By looking at prediction-market data, they find that changes in the probability that the incumbent party gets re-elected are the largest determinant of financial market uncertainty, not simply the presence of an election. Goodell et al. credit this to the changes that arise when new administrations enter office: this finding can be summarized by the well-known saying that the devil you know is better than the one you don't. They also find that uncertainty increases as an election gets closer, when measuring the eight weeks leading up to an election. This finding specifically has a large implication on this paper; a dummy variable that treats every day in an election year identically may be too broad to capture the uncertainty they found.

Finally, the CME Group/Coalition Greenwich (2025) industry survey offers a market perspective on the importance of Treasury volatility. It finds that 81% of participants monitor the metric multiple times a day and that 83% trade Treasury volatility to manage risk within portfolios. This survey is useful to show how volatility specifically is monitored often by market participants, reinforcing the idea to model volatility rather than looking at only yields.

When looking at these four sources, they suggest that Treasury yield volatility is largely driven by macroeconomic conditions and news regarding the business cycle and that yields respond with a lag. They also suggest that political uncertainty does not affect markets equally across the business cycle, or across years containing elections. None of these sources directly measure whether a presidential election has a noticeable effect on volatility while controlling for macroeconomic conditions. This study seeks to

fill this gap using GJR-GARCH (1,1) volatility modeling, two-sample t-tests and probit models for all three Treasury tenors.

Data

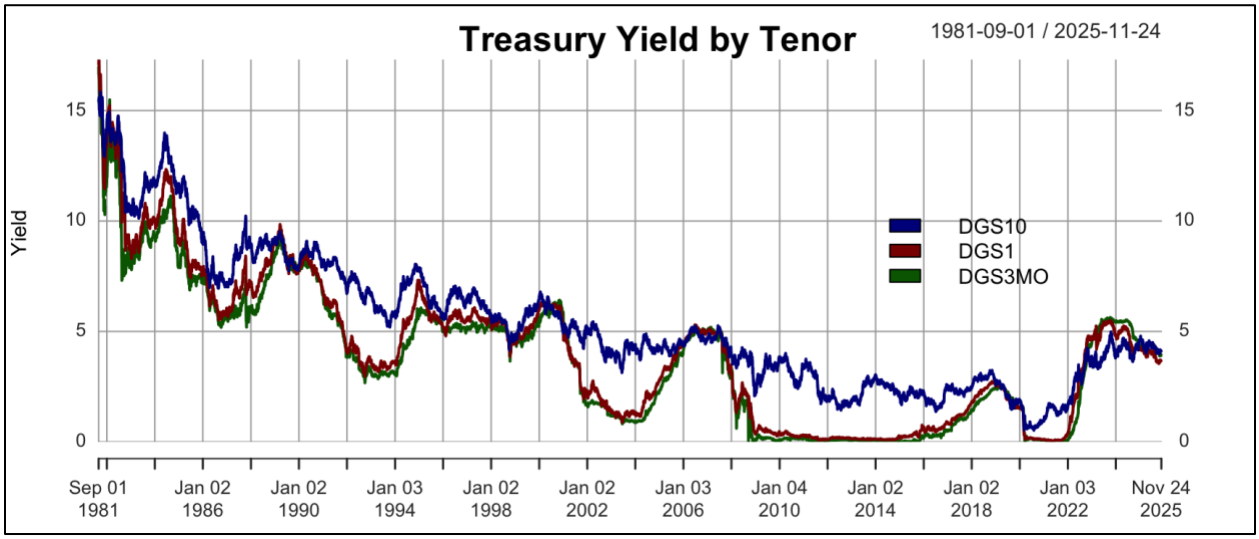


Figure 1: U.S. Treasury Yields by Maturity, Sep. 1981–Nov. 2025

All data come from FRED. I use daily yield series for the 3-month (DGS3MO), 1-year (DGS1), and 10-year (DGS10) Treasury, alongside the NBER recession indicator (USREC) and the USD/EUR exchange rate (DEXUSEU) as controls. Daily returns were calculated from each yield series by taking the difference between period *t* and *t-1*. Observations were aligned across the sample period of September 1981 through November 2025 to ensure comparability across tenors totaling 11,057 daily return observations per series[1].

[1] The 3-month return series contains one fewer observation (11,056) than the 1-year and 10-year series, reflecting a later start for the underlying 3-month yield data.

Table 1: Descriptive Statistics of Daily Returns					
Tenor	**N**	**Mean**	**Std. Dev.**	**Skewness**	**Kurtosis**
3-Month	11,056	-0.0012	0.0725	-0.0039	54.79
1-Year	11,057	-0.0012	0.0655	-0.7932	27.38
10-Year	11,057	-0.0010	0.0667	-0.2546	5.589

Descriptive statistics reveal that all three-return series are heavy-tailed, with the 3-month return series showing very high excess kurtosis (54.8), the 1-year series showing similarly elevated kurtosis (27.4), and the 10-year series showing comparatively moderate excess kurtosis (5.6)[2]. Fatter tails at the short end of the curve are an early signal that short-term Treasuries are more prone to sharp, outlier-driven moves than long-term Treasuries, independent of any election effect.

I constructed a binary indicator for each tenor equal to 1 when a daily return exceeded two standard deviations in magnitude, titled more2vol, more2vol1, and more2vol3 which correspond to the 10-year, 1-year, and 3-month Treasuries respectively[3]. This serves as the dependent variable in both the two-sample t-tests and the probit regressions described below. I also constructed a binary election-year indicator equal to 1 for each tenor, indicating a year containing a U.S. presidential election[4].

Methodology

To determine the probability of a binary outcome, a probit model is used. Since these coefficients are uninterpretable on their own, marginal effects must also be applied to recognize outcomes. The structure of the model, at time *t*, is as follows:

[2] The package "Fbasics" in R reveals excess kurtosis, not total kurtosis. While some packages may return total kurtosis, requiring a manual subtraction of 3 to find excess, this package automatically does that.

[3] The 10-year volatility indicator is named more2vol (without a tenor suffix) since it was the first constructed in the analysis; the 1-year and 3-month indicators were named more2vol1 and more2vol3 respectively to distinguish them once the analysis expanded to three tenors.

[4] Creating three election dummy variables was done to ensure homogenous election years across the three tenors. Since the 3-month yield's observations started much later than the others, there were potential indexing issues, and this was the fix at the time. This would be improved in a future expansion of the project.

$$P(more2vol_{i,t} = 1) = \Phi(\beta 0 + \beta 1 \cdot election_dummy_t + \beta 2 \cdot USRec_t + \beta 3 \cdot DEXUSEU_t) \quad (1)$$

This model measures the probability of more than two standard deviations occurring when controlling for election years, recessions and exchange rates between the USD and EUR. These macroeconomic variables were included as the existing literature revealed that Treasury yields react more to economic conditions and news than they do to political uncertainty. The exception to this is when the economy is running poorly, then political uncertainty has a larger effect on asset valuations. In this equation $more2vol_{i,t}$ is the binary variable indicative of greater than two standard deviations at time *t* and for the tenor, *i,* and $election_dummy_t$ indicates if *t* is in an election year[5]. $USRec_t$ indicates whether time *t* is during a recession and $DEXUSEU_t$ measures the nominal exchange rate at time *t.* Finally, Φ represents the standard normal cumulative distribution as a probit is used and not a logit, which would require a logistic distribution.

To forecast volatility using a GJR-GARCH model, the data must have certain characteristics, the first being stationarity. When observing each of the Treasury yields it is clear this is not the case, however, to ensure yields are not stationary, an Augmented Dickey-Fuller test (ADF) is used. This test's

Table 2: Stationarity Test: Augmented Dickey Fuller Test p-values (Yield and Return)

Yield		Return	
Tenor	**Value**	**Tenor**	**Value**
3-Month	0.08	3-Month	< 0.01
1-Year	0.045	1-Year	< 0.01
10-Year	0.038	10-Year	< 0.01

[5] Where *i=3, more2vol3* is the dependent variable, where *i=1 more2vol1* is the dependent variable, where i=10 more2vol is the dependent variable.

null hypothesis is non-stationarity and each tenor's yield produced a p-value confirming this, or just barely refuting it; returns showed much more stationarity when assessing their P-values.

Pairing the ADF tests with the visual analysis of yields compared to returns, the graphs of both reveal their true nature: returns seem to exhibit mean reverting qualities while yields closer resemble a variable with a unit root[6]. These tests show that returns would be prudent to use when forecasting Treasury volatility. Additionally, a two-sample t-test is used to cross check if there is a difference in mean returns between election and non-election years, simply to verify or refute the probit results.

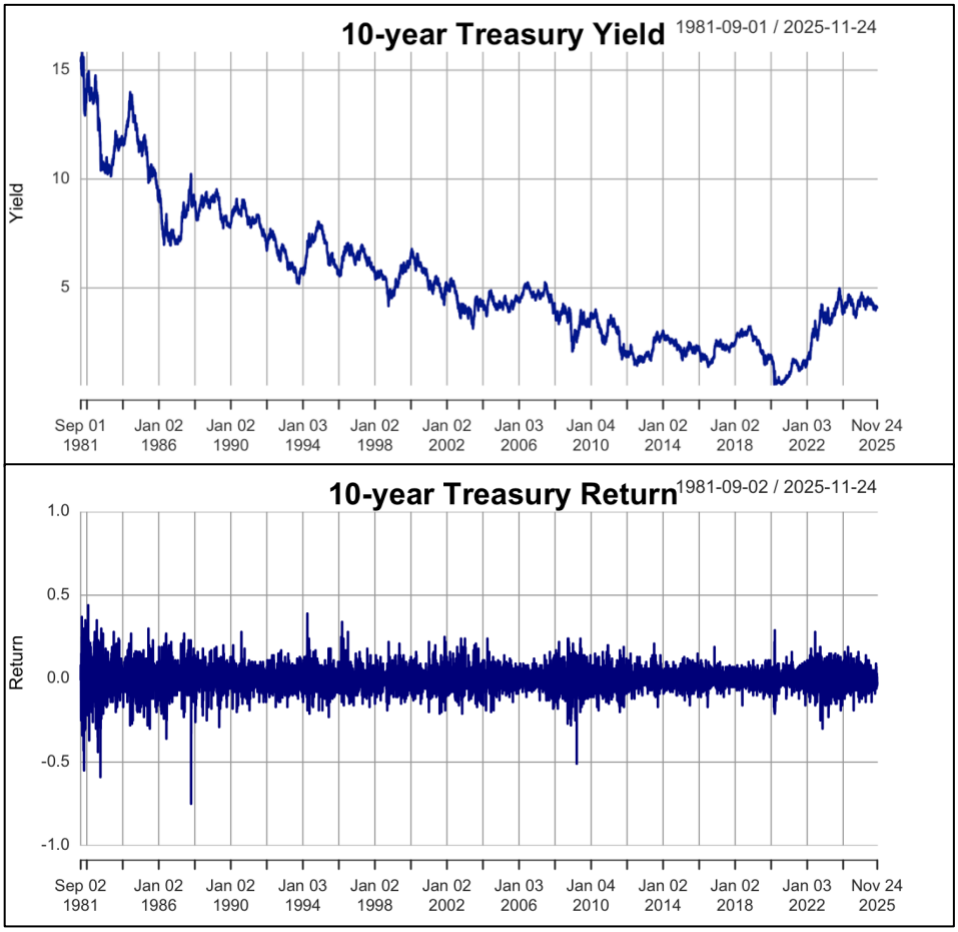


Figure 2: 10-Year Treasury Yield and Daily Return, Sep. 1981–Nov. 2025

[6] See Appendix for 3-month and 1-year

To forecast volatility, a GJR-GARCH (1,1) model is used. Specifically for this research the following equation is utilized:

$$\sigma_{i,t}^2 = \omega_i + \alpha_i r_{i,t-1}^2 + \gamma_i r_{i,t-1}^2 \cdot \mathrm{I}(r_{i,t-1} < 0) + \beta_i \sigma_{i,t-1}^2 \qquad (2)$$

For this model, $\sigma_{i,t}^2$ represents the daily variance at time *t* and $r_{i,t-1}^2$ represents the previous days squared returns, to ensure that they are positive[7]. For the baseline GARCH parameters (ω_i, α_i, β_i) ω_i represents a baseline daily variance for a given tenor, *i*. α_i represents a sensitivity to previous days squared returns, and β_i represents how much of time *t*'s volatility is based on the previous period's volatility. As negative returns tend to have higher effects on an asset's volatility, $\mathrm{I}(r_{i,t-1} < 0)$ indicates whether a return is negative and γ_i measures the added effect that previous negative returns have on variance in the current period. This forecasting formula was applied to the 10-year and 1-year Treasuries, however the 3-month tenor experienced extremely high kurtosis and often suffered from high outliers, making forecasts unreliably biased from large shocks.

The following equation was used to turn daily variance into daily standard deviation, or volatility:

$$\sigma_{i,t}^{daily} = \sigma_{i,t} = \sqrt{\sigma_{i,t}^2} \qquad (3)$$

Once daily volatility was found, it was annualized to reflect yearly volatility:

$$\hat{\sigma}_{i,t}^{annual} = \sigma_{i,t}^{daily} \cdot \sqrt{252} \qquad (4)$$

Combining all these equations provided the information necessary to determine whether presidential elections had any effect on Treasury Volatility, as well as forecast volatility 365 days forward using the GJR-GARCH model.

[7] GARCH models use squared returns to ensure volatility is a positive number: positive and negative swings may cancel each other out, but squaring returns result in positive values always. The GJR-GARCH model accounts for this by having a separate variable which accounts for negative returns, γ_i.

Results

Running the probit regression for each of the Treasury tenors and then applying the marginal effects to get interpretable coefficients provides the following results:

Table 3: Election Year Effects on Probability of Extreme Volatility

Tenor	Election Marginal Effect	% Increase (-)	p-value
3-Month	0.069	7%	0.158
1-Year	-0.014	(-1%)	0.684
10-Year	-0.051	(- 5%)	0.041*

For the 3-month securities, despite statistical insignificance (p-value = 0.158), days in election years correspond to a noticeable increase in their probability of high volatility, by roughly 7%. For 1-year Treasury securities, there is a very small decrease in the probability of high volatility, around a 1% decrease, despite statistical insignificance (p-value = 0.684). The only tenor of security that shows any statistically significant result (p-value = 0.041) is the 10-year Treasury; these securities experienced around a 5% decrease in the likelihood of experiencing high volatility during presidential election years. This finding contradicts the findings in the research conducted by Goodell et al. (2020) as they find that elections caused heightened uncertainty, as the incumbent party's odds of re-election change and political change looms; perhaps the method in this paper, of creating a dummy variable that treats every day in an election year identically, may be too coarse to reinforce their research. One explanation for the 10-year's decrease in volatility, using the logic of this previous paper, is that incumbent presidents' probabilities of re-election have themselves not changed much over the years observed, ultimately lowering volatility in financial markets. This explanation is consistent with this paper's findings and those by Goodell et al.

It is important to observe the other variables in each probit regression as well; Table 4 shows simplifications for each of the three probit regressions.

Table 4: Recession and Exchange Rate Effects on Probability of Extreme Volatility

Tenor	Recession Marginal Effect	%	Recession p-value	Exchange Rate Marginal Effect	%	Exchange Rate p-value
3-Month	-0.0281	-2.81%	0.569	-3.384	-338%	0.199
1-Year	0.071	7.1%	0.406	-2.581	-258%	0.273
10-Year	0.127	12.7%	0.216	-0.182	-18%	0.932

Before interpreting the results of the remaining variables, it is important to clarify the nature of them, specifically the exchange rate effect. The recession indicator is binary (1 signifying recession, 0 otherwise) but the exchange rate is simply a level, not the daily difference; this differentiation skews the interpretation of the exchange rate coefficients to the point of economic triviality. Since the exchange rate is measured as a level, the dollar would have to appreciate one hundred percent relative to the euro for the shown decreases to materialize, an occurrence that itself is unrealistic. The sign of these coefficients is the only salvageable finding, and they show that a higher value of the dollar relative to the euro causes less probability of volatility; a properly specified model, which uses the difference rather than level, would provide more meaningful insights into the relationship between exchange rates and Treasury volatility.

Despite high statistical insignificance, the effects of recessions on Treasury volatility seems to echo the findings of Pástor and Veronesi (2013) and how volatility is greater in worse economic conditions. 10-year and 1-year Treasuries respond to recessions with higher volatility, while 3-month securities see a decreased chance of experiencing high volatility. Since 3-month securities have extremely high kurtosis (see Table 1) the unique effect recessions have can be attributed to random chance as opposed to a definitive causal relationship. 1-year and 10-year securities seem to experience more sizeable, directionally sound, swings in probability of experiencing high volatility (7.1% and 12.7% respectively).

Two-sample t-tests assess whether the frequency of high-volatility days differs significantly between election years and non-election years; they offer the same results as the probit regression.

Table 5: Two-Sample t-test Measuring Probability of Elevated Volatility Between Election and Non-election years

Tenor	t-statistic	p-value
3-Month	-1.265	0.2097
1-Year	0.078	0.9376
10-Year	1.506	0.134

The findings in this paper reveal that there is no statistical difference in the percent chance of experiencing high volatility between election years and non-election years, as all the p-values are noticeably higher than 5% or 10% at the highest bound. These findings reinforce the research conducted by Moench & Soofi-Siavash (2022), ultimately finding that presidential elections do not move Treasury markets as much as the initial hypothesis believed.

In addition to testing for an election effect, the GJR-GARCH(1,1) models were also used to forecast annualized volatility 365 days from the end of the sample period for the 10-year and 1-year Treasuries, Table 6 shows estimated values for each of the specific parameters for this model.

Table 6: GJR-GARCH(1,1) Parameter Estimates: 10-Year and 1-Year Treasuries

Parameter	10-Year Estimate	1-Year Estimate
Baseline Volatility - ω	0.000000	0.000001
Sensitivity to Shock - α	0.0494	0.0658
Persistence of Prior Volatility - β	0.9461	0.9164
Additional Volatility from Negative shocks - γ	0.0070	0.0336

Part of the difference in volatility between the 10-year and 1-year Treasuries come from their differing values of α and γ. Volatility for the 10-year relies less on prior period shocks than the 1-year, despite having more persistent elevated volatility. Additionally, the 1-year is more susceptible to volatility stemming from previous negative returns, nearly five times that of the 10-year.

Forecasted annualized volatility remains less than 1% across the forecast period for the 10-year Treasury: the value at the end of the period is 0.93%, which is shown in the red line in Figure 3.

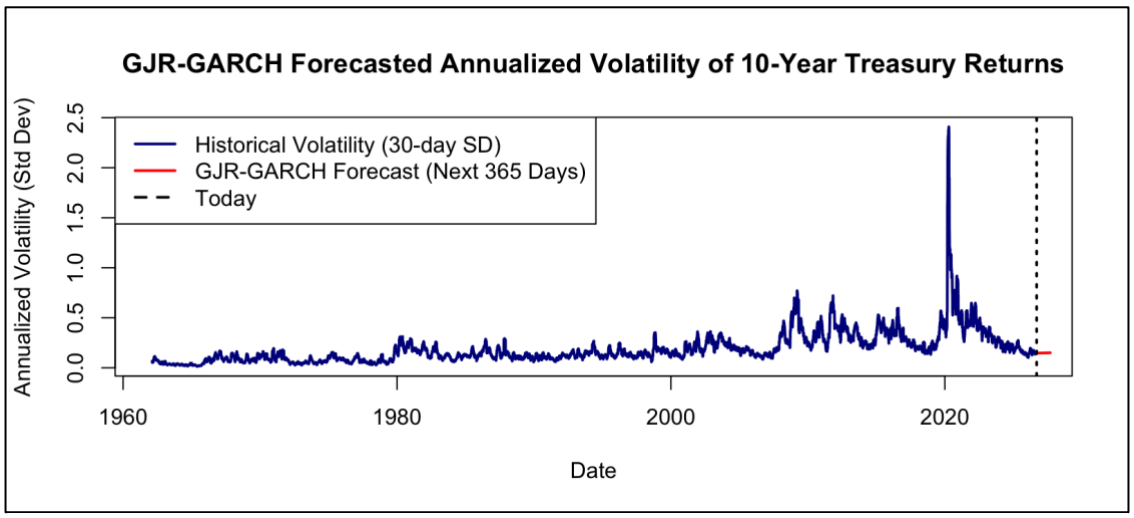


Figure 3: GJR-GARCH(1,1) Annualized Volatility Forecast: 10-Year Treasury

For 1-year Treasuries, annualized volatility was around 2% (1.93%) which is still low, but higher than that of the 10-year: this can be seen with a steeper forecast than the 10-year, shown in green.

This finding is consistent with the conclusion of Moench & Soofi-Siavash (2022), which shows how a larger portion of Treasury yield variation is driven by shocks whose effects compound and are not resolved immediately. The estimates α and γ explain why longer dated securities have lower volatility than their shorter counterparts: longer dated securities are less reactant to short developments such as large or negative returns. The 10-year has more inertia in its volatility than the 1-year, as high volatility persists for more time despite experiencing spikes less often. Ultimately, longer maturity securities are less influenced by short-term developments and more by secular trends, a finding consistent with previous literature.

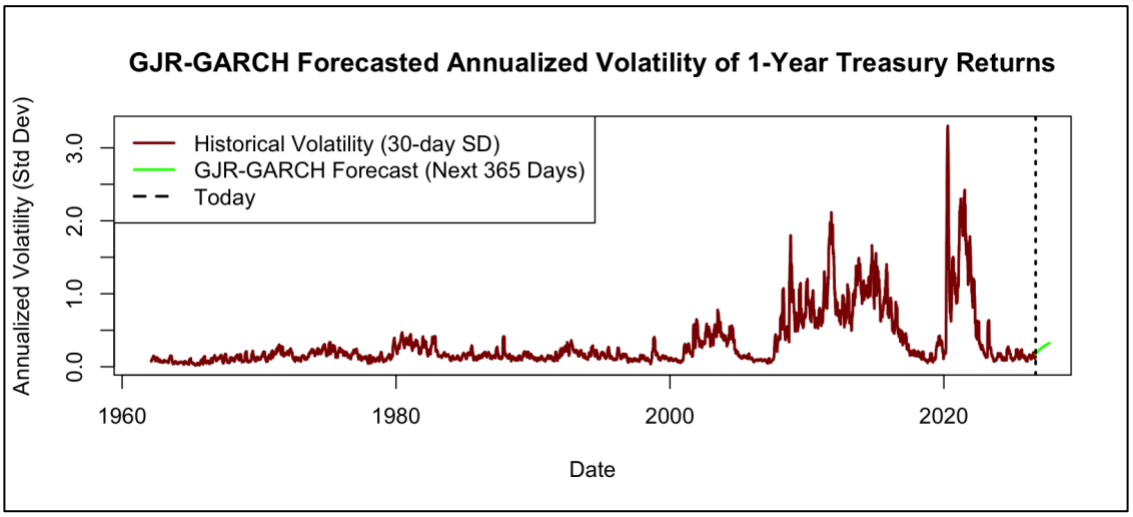


Figure 4: GJR-GARCH(1,1) Annualized Volatility Forecast: 1-Year Treasury

Conclusion

When understanding these results in their entirety, there is no evidence to support the hypothesis that Treasury securities become more volatile due to uncertainty regarding presidential elections. The two-sample t-tests show little difference between probability of volatility in election years compared to non-election years across all tenors. Probit regressions for 3-month and 1-year Treasuries produced statistically insignificant results, reinforcing the claims of Moench and Soofi-Siavash (2022) that volatility largely results from negative business cycle news. Forecast parameter estimates for the GJR-GARCH(1,1) model also seem to explain why different tenors experience volatility differently: new shocks and negative returns impact shorter securities more than they do longer securities, while longer securities experience more persistent volatility. The probit for 10-year volatility produced a statistically significant result opposite to the original hypothesis, finding that elections decreased volatility. Based on the findings of Goodell, McGee, and McGroarty (2020) one explanation for this may be that incumbent parties in this data experienced relatively stable re-election odds, which they find is the largest driver of election-related volatility.

Future research can address the limitations that arose from the exchange rate variable's form and measuring an election year identically across all its days. Existing literature finds that there is a relationship between political uncertainty and financial markets; differencing the exchange rate and shortening the effective window of election uncertainty could further reinforce these works and strengthen the findings of this paper. In the future, amending these variables to better capture any effects, as well as controlling for macroeconomic news and potential Federal Reserve actions, may produce more statistically sound results.

The main takeaway is that Treasury volatility does not seem to be influenced materially by election cycles; divesting from these securities during presidential election years may be riskier than waiting out any short-term developments. Treasury securities maintain their status as a safe way to generate return for both institutional and retail investors alike.

References

Goodell, J. W., McGee, R. J., & McGroarty, F. (2020). Election uncertainty, economic policy uncertainty and financial market uncertainty: A prediction market analysis. *Journal of Banking & Finance*, *110*, 105684. https://doi.org/10.1016/j.jbankfin.2019.105684

Liang, N. (2025, April 14). What's going on in the US Treasury market, and why does it matter? *Brookings*. https://www.brookings.edu/articles/whats-going-on-in-the-us-treasury-market-and-why-does-it-matter/

Moench, E., & Soofi-Siavash, S. (2022). What moves treasury yields? *Journal of Financial Economics*, *146*(3), 1016–1043. https://doi.org/10.1016/j.jfineco.2022.04.001

Pástor, Ľ., & Veronesi, P. (2013). Political uncertainty and risk premia. *Journal of Financial Economics*, *110*(3), 520–545. https://doi.org/10.1016/j.jfineco.2013.08.007

Understanding the importance of measuring U.S. Treasury volatility (Coalition Greenwich Report). (2025, July 22). CME Group. https://www.cmegroup.com/articles/whitepapers/understanding-the-importance-of-measuring-us-treasury-volatility.html

Appendix

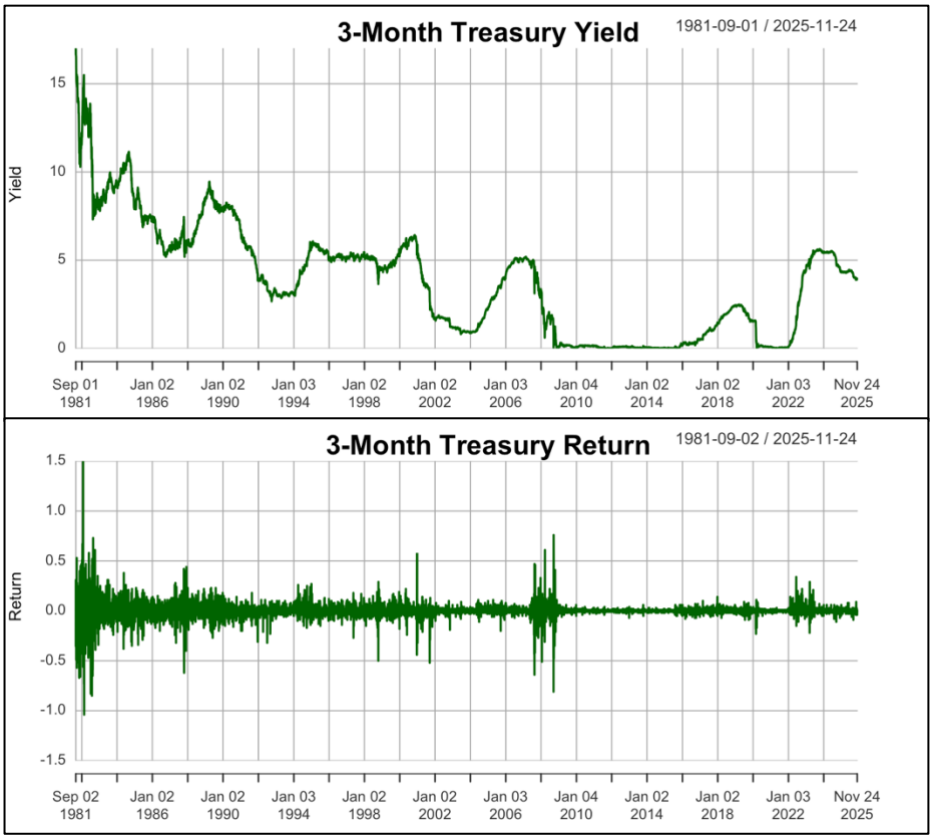


Figure A-1: 3-Month Treasury Yield and Daily Return, Sep. 1981–Nov. 2025

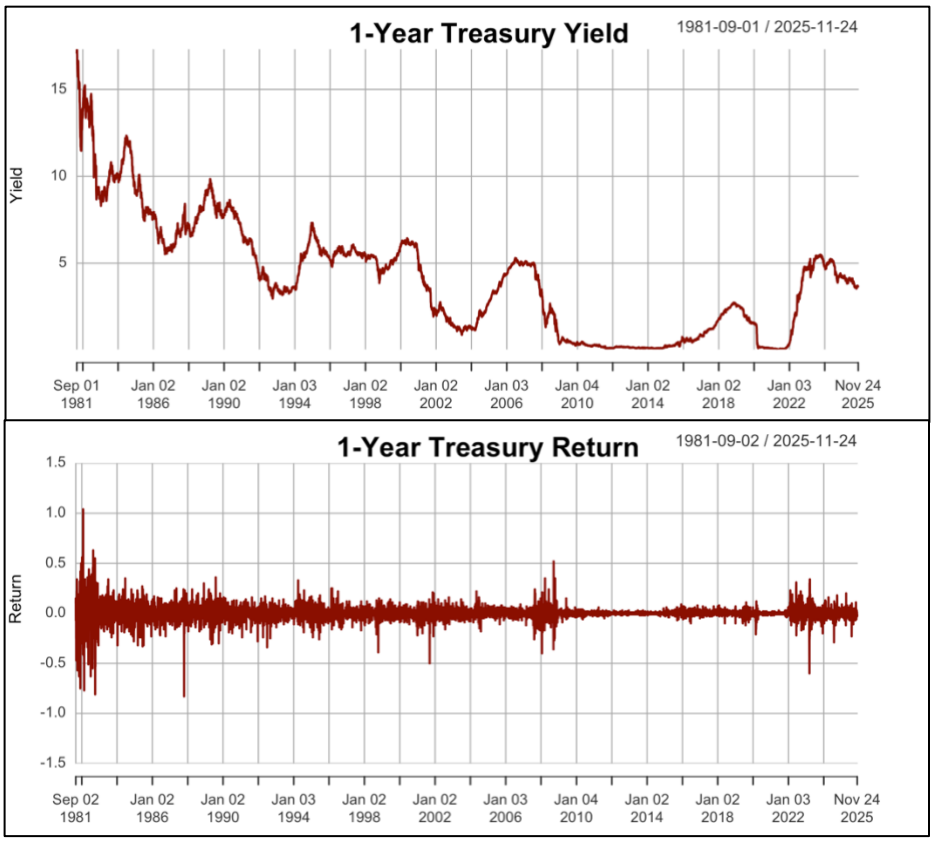


Figure A-2: 1-Year Treasury Yield and Daily Return, Sep. 1981–Nov. 2025